

data validation, Security & password, Subtotal

① How do you create a drop list using data validation in Excel?

* A drop-down list in excel is created using data validation to restrict data entry.

First, select the cell where the list is required.

Then go to data → data validation

In the Allow box, choose list and enter the values ~~spe~~ separated by commas or select a cell range containing the list items.

② What steps would you take to ensure data validation rules are applied correctly?

* To ensure data validation rules work correctly, first select the correct cells before applying the rule.

use clear validation criteria and appropriate error ~~set~~ alerts.

Test the validation by entering both valid and invalid data.

use input message to guide users and error alerts to prevent incorrect entries.

③ Explain the process of protecting a worksheet with a password?

→ To protect a worksheet, go to the Review tab and click Sheet.

Choose the actions users are allowed to perform, such as selecting cells, enter a password and confirm it.

Once protected, users cannot modify locked cells without the password.

This helps prevent unauthorized changes to the worksheet.

④ How do you protect specific cells while allowing others to be edited?

→ By default, all cells are locked in Excel. First, select the cells that should remain editable and open Format Cells → Protection.

uncheck locked and click OK.

Then protect the sheet using Review → Protect Sheet with a password.

Only locked cells will be protected; unlocked cells remain editable.

⑤ what is the purpose of the SUBTOTAL function and how is it different from SUM?

→ The SUBTOTAL function performs calculations like SUM, average or count on filtered data.

unlike the SUM function, SUBTOTAL ignores hidden rows created by filters.

It is useful in dynamic data analysis where filters are frequently applied.

SUBTOTAL also avoids double counting when used with nested subtotals.

⑥ How do you apply data validation to prevent duplicate entries?

→ To prevent duplicate entries, select the target cells and open data validation.

Choose custom and enter a formula such as

$$\text{=COUNTIF}(A:A, A1)=1.$$

Set an appropriate error message to inform users.

This ensures only unique values are entered in the selected range.

It is commonly used for ID, serial numbers, or invoice numbers.

⑦ How would you restrict data entry to a specific range of values?

→ Select the cells and go to data → data validation
choose whole number or decimal under allow
Specify the minimum and maximum values allowed.

Add an error alert to display a message if invalid data is entered.

This ensures accurate and controlled data input.

⑧ What are the security options available in excel for protecting workbooks?

→ Excel provides several security options such as protect sheet, protect workbook, and encrypt with password.

You can restrict structure changes like adding or deleting sheets.

Passwords can be set to open or modify a file.

Workbook sharing and permission can also be controlled.

These options help maintain data confidentiality and integrity.

9) How do you hide formulas in a worksheet while still allowing data entry?

* To hide formulas, select the formula cells and open format cells → protection. check Hidden and ensure the cells are locked. Then protect the worksheet using protect sheet.

Formulas will not be visible in the formula bar, but users can still enter data in unlocked cells.

This protects calculations from being altered

10) Explain how to use the SUBTOTAL function to perform calculations on filtered data.

* The SUBTOTAL function is used to calculate values like sum or average on filtered data.

Syntax: =SUBTOTAL (function_num, range)

For example, =SUBTOTAL (9, A2:A20)

Calculate the sum of visible row only.

When filters are applied, hidden rows are automatically ignored.

This makes SUBTOTAL ideal for analyzing filtered lists and tables.